Delhi Technological University

B. Tech, Third Semester

END SEMESTER EXAMINATION November 2013

CE/EE/EP/EL/MC/SE-206-Engineering Economics

Time: 3 Hour Max. Marks:70

Note: Answer any five questions,
Assume suitable missing data, if any
Use of interest rate table is allowed

1.	a	What do you mean by Business Cycle? Discuss the phase of the Business cycle through which the Indian Economy is undergoing at present.	7
	b	Find the price elasticity of demand for the demand function $P = e^x$ and $P = e^x/x$	4
	c	If TC = ax2 +bx +c is the cost function of a monopolist and $P = \beta + \alpha x$ is the demand law, find the monopoly price and output.	3
2.	a	Discuss role of appropriate technology in the development of a developing economy.	7
	b	A person who is now 35 years old is planning for his retired life. He plans to invest an equal sum of Rs. 10,000 at the end of every year for the next 25 years starting from the end of the next year. The bank gives 20% interest rate, compounded annually. Find the maturity value of his account when he will be 60 years old.	4
	c	A company is producing tire. Cost of machine is Rs. 261,000. Cost of labour and raw material is @Rs. 160/tire. Manufacturer sells tire @ Rs.450/tire. What will be the break even point for the company?	3
3.	a	What is the significance of studying Engineering Economics for engineering students? How it can be mad more relevant.	7
	b	Two years ago, a machine was purchased at a cost of Rs. 2,00,000 to be useful for eight years. Its salvage value at the end of its life is Rs. 25,000. The annual maintenance cost is Rs. 25,000. The market value of the present machine is Rs. 1,20,000. Now, a new machine to cater to the need of the present machine is available at Rs. 1,50,000 to be useful for six years. Its annual maintenance cost is Rs. 14,000. The salvage value of the new machine is Rs. 20,000. Using an interest rate of 12%, find whether it is worth replacing the present machine with the new machine.	7

4. a Discuss factors which should considered while deciding price of the product in the market.

b Rs. 40,000 were taken as loan for replacement of a machine and it compounds to Rs. 60,000 in 5 years, what will be the rate of interest which is compounded quarterly.

c Following are the demand functions for two commodities x_1 and x_2:

$$x_1 = p_1^{-1.7} p_2^{0.8}$$
$$x_2 = p_1^{0.5} p_2^{-0.2}$$

Determine whether the commodities are complementary or competitive.

5. a Why do businesses such as automobile dealers tend to locate together? Three engineering graduate decides to become automobile dealer, and all three devote their full time to its management. What opportunity cost would you assign to their time?

b A company must decide whether to buy Machine A or Machine B be Future Worth method. Rate of interest 12% which is compounded annually.

	Machine A	Machine B
Initial Cost in Rs.	4,00,000	8,00,000
Useful life in years	4	4
Salvage value at the end of machine life in Rs.	2,00,000	5,00,000
Annual maintenance Cost in Rs.	40,000	NIL

6. a Discuss the nature of Computer Software market in India. What incentive may be given by the Government to encourage Indian producer?

b. For construction purpose, a concrete aggregate mix is required which should contain at least 34% sand by volume. We have following option available:

	Sand	Coarse	Rate
One source material	30%	70%	Rs. 3.50 per cubic meter
Second source material	42%	58%	Rs. 4.25 per cubic meter

Determine the least cost per cubic meter of blended aggregates.

c. A special type of screw is manufactured at a unit cost of Rs. 4 for material and Rs. 5.5 for direct labour. An investment of Rs. 6,00,000 is required to purchase the machine. Other cost is 1.5 times Direct labour cost. The order calls for three million pieces. Half-way through the order, a new method of manufacture can be put into effect which will reduce the unit costs of Rs.3

for material and Rs. 4.5 for direct labour –but it will require Rs1,00,000.00 for additional machine. Which process should be taken up?

7. a Discuss seven salient feature of the Indian Economy. 7

b The green houses have the following cost comparison during their life time production system. Find out which system is more economical if the rate of interest is 15% compounded annually. 7

	System A	System B
Initial Cost in Rs.	30,000	20,000
Maintenance Cost/year in Rs.	1000	2500
Replacement of a component at the end of every third year in Rs.	4000	4200
Salvage value in Rs.	5,000	-
Life in Years	20	20
Benefit from quality control as uniform end of year amount/year	1000	-